

A Method for Resolving Overlapped Peaks in Laser-Induced Breakdown Spectroscopy (LIBS)

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Spectral peak overlapping is a basic problem in analytical data processing of laser-induced breakdown spectroscopy (LIBS). Curve fitting is the typical method of resolving overlapped peaks. For preventing ambiguous fitting, appropriate initial values must be known. The aim of this work was to present a method that could be used to determine appropriate initial values of the curve-fitting method by using fractional differential theory. According to the variation of characteristic points of Lorentzian peaks at different fractional differential orders, parameter estimators were obtained that were used to calculate the initial values of the curve-fitting method. As it is a widely used optimization method, the Levenberg–Marquardt method was used in curve fitting. Simulation and LIBS experimental results proved that the proposed method of the initial value estimation can effectively resolve the overlapped peaks in LIBS data processing.

Index Headings: Overlapped peaks; Laser-induced breakdown spectroscopy; LIBS; LIBS data processing; Curve fitting; Fractional differential.

INTRODUCTION

Laser-induced breakdown spectroscopy (LIBS) applications for the analysis of elemental composition of materials in solid, liquid, and gaseous states have been extensively studied and reviewed.^{1–4} Peak overlapping commonly occurs in LIBS data. Although improving the performance of experimental equipment should reduce the interference of overlapped peaks, the high price for hardware with limited resolving effect may be unsatisfactory. Signal-processing methods such as the Fourier deconvolution method, wavelet method, neural network method, and curve-fitting method^{5–8} have become increasingly important to improve the resolution of overlapped peaks. However, all of these methods have different problems. The change of width in resolved overlapped peaks by using the Fourier deconvolution method causes inaccurate results. For the wavelet method, there are no uniform rules to determine the decomposition levels and wavelet basis functions, leading to the need for many comparison experiments with different decomposition levels and wavelet

basis functions. For the neural network method, the approaching ability and generalization performance are correlated with the representativeness of learning samples; however, the selection of the training set in typical samples is very difficult. In the curve-fitting method, some prior information is required, such as the number of individual peaks in overlapped peaks and the reasonable selection of initial values for optimization methods.^{9,10}

Usually, the curve-fitting method for resolving overlapped peaks in LIBS data has three key procedures: (i) The selection of peak mathematical models; the Lorentzian shape is commonly predicted by LIBS physics.¹¹ (ii) The selection of optimization methods for the best approximation between the model and the experimental data. The global search technique for curve fitting based on evolutionary random search is modified and applied for quantifying a combination of Lorentzian peaks.^{12,13} The Levenberg–Marquardt method combines the Gauss–Newton method and the Steepest decent method, as it has quick convergence and global search characteristics.¹⁴ (iii) The determination of the initial values for optimization methods. The convergence speed, computation time, and resolving effect are extremely affected by the initial values that must be accurately estimated in advance. Unfortunately, the initial values are usually selected based on operator experience, which contributes to the unreliable results of optimization methods. Basically, the selection of peak mathematical models and optimization methods is confirmable for the curve-fitting method. However, the selection of initial values, which directly affects the accuracy and reliability of the curve-fitting method, is uncertain.

Here, a method for the initial value estimation is proposed by using the variation of the characteristic points, extremum and zero-crossing points, of the peak model at different fractional differential orders.¹⁵ After the determination of the initial values, the Levenberg–Marquardt method was selected for the best approximation between the peak model and the experimental data. To demonstrate the feasibility of the method, we successfully resolved overlapped peaks of synthetic and experimental LIBS data by using the method.

THEORY

Peak Models. The Voigt function might be preferred to fit peaks accurately. However, we use the Lorentzian function to

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fit peaks for simplicity in this study, not only because the Voigt function and the Lorentzian function return the same peak positions but also because the Voigt function requires an additional parameter. In practical applications, the overlapped peaks are usually composed of several individual peaks. The individual peaks can be described by a Lorentzian function that contains three parameters: peak height, position, and width. The noise in the overlapped peaks is assumed to be filtered, and the overlapped peaks are expressed as follows:

$$F(x) = \sum_{j=1}^M L_j(x; A_j, \mu_j, \sigma_j) = \sum_{j=1}^M A_j \cdot \frac{\sigma_j^2}{(x - \mu_j)^2 + \sigma_j^2} \quad (1)$$

where $L_j(x; A_j, \mu_j, \sigma_j)$ is the individual peak with the wavelength variable x ; A_j , μ_j , and σ_j are the peak height, peak position, and full width half-maximum (FWHM), respectively; and M and j are the number of the individual peaks and the serial number of the individual peaks, respectively. The curve-fitting method, also known as nonlinear regression, is a mathematical tool for modeling the experimental data. The adjustable parameters of the model are varied until the best agreement between the data and the model is obtained. The least squares solution is obtained by minimizing the objective function described by Eq. (2):

$$RSS = \left(\sum_{i=1}^N y(x_i) - F(x) \right)^2 \quad (2)$$

where the experimental data as a function of the variable x is given by $y(x_i)$, N gives the number of the experimental data, and the fitting data are given by $F(x)$. For obtaining the best-fitting parameters, a reasonable selection of the initial values is necessary.^{16,17} Improper selection of the initial values will result in divergence and more iterations. It is worth mentioning that an offset (a constant background) also was included in the curve-fitting process.

Determination of the Initial Values in Curve Fitting. The initial values in curve fitting include the number, positions, heights, and FWHMs of the individual peaks. Note that the curve-fitting method is mathematically non-unique under an acceptable error. One of the main drawbacks is that as the initial parameters of individual peaks are different, the results may become progressively ill conditioned. Thus, any overlapped peaks may be fitted by different groups of the individual peaks. Fortunately, this drawback can be avoided by determining the appropriate initial values that includes the number, positions, heights, and FWHMs of the individual peaks.

There are many ways to determine the number and positions of the individual peaks, such as the higher order differential method, factor analysis, the Fourier self-deconvolution method, and the wavelet method.^{18–21} However, the way to set the heights and FWHMs of the individual peaks is based on operator experience, which induces poor convergences. In this study, continuous spline wavelet transform (CSWT) was used to determine the number and positions of the individual peaks. To effectively determine heights and FWHMs of the individual peaks, we used a method based on the fractional differential theory developed by Leibnitz. The fractional differential theory provides the information of interpolation between integer order differential. This interpolation can be useful when studying the behavior of characteristic points, such as extreme values and zero-crossing

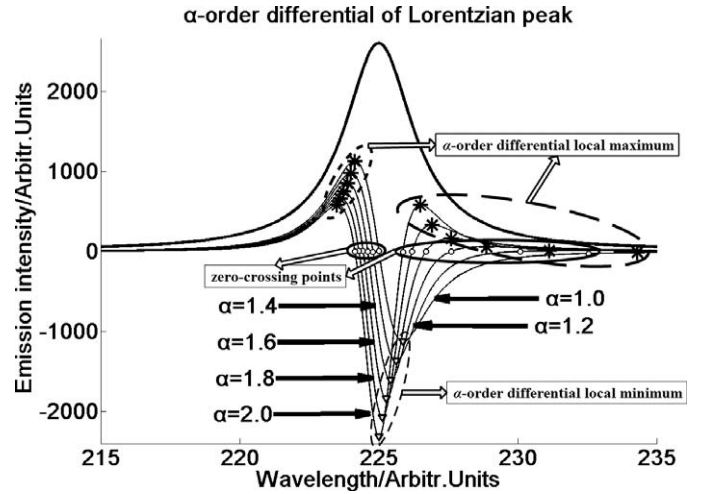


Fig. 1. Fractional differential of Lorentzian peak when $A = 2600.0$, $\mu = 225.0$, $\sigma = 1.5$, and $x \in [215 \ 235]$.

points. The three most frequently used definition for the general fractional differintegral are as follows: the Grünwald–Letnikov (GL) definition, the Riemann–Liouville definition, and the Caputo definition.^{22–24} Here, the GL definition was used to define the α -order differential of the signal. The α -order differential of signal $f(t)$ is defined as follows:

$$df(t) = h^{-\alpha} \sum_{j=0}^{\lfloor \frac{t-d}{h} \rfloor} d_j^\alpha f(t-jh) \quad (3)$$

where

$$d_j^\alpha = \begin{cases} 1, & j = 0 \\ (-1)^j \frac{\alpha(\alpha-1)(\alpha-2) \dots (\alpha-j+1)}{j!}, & j > 0. \end{cases}$$

The upper and lower limit of differentiation is given by t and d , respectively. The sampling period and order of differentiation are represented by h and α . $\lfloor (t-d)/h \rfloor$ is integer part of $(t-d)/h$. In addition, the first-order differential and second-order differential of the Lorentzian function are given by Eq. (4):

$$\begin{aligned} L^{(1)}(x) &= -2A\sigma^2 \cdot \frac{(x - \mu)}{[(x - \mu)^2 + \sigma^2]^2}, \\ L^{(2)}(x) &= 2A\sigma^2 \cdot \frac{3(x - \mu)^2 - \sigma^2}{[(x - \mu)^2 + \sigma^2]^3} \end{aligned} \quad (4)$$

As an example, the α -order differential of the Lorentzian peak is shown in Fig. 1. We can see that the extremum and zero-crossing points vary as the differential order α . Moreover, the multiset of α -order differential local maximum and zero-crossing points is demarcated by the α -order differential local minimum. After analysis of the behavior of the characteristic points, a quadratic polynomial was used to fit the relationship between the characteristic points and its corresponding differential order:

$$x(\alpha) = a_2\alpha^2 + a_1\alpha + a_0 \quad (5)$$

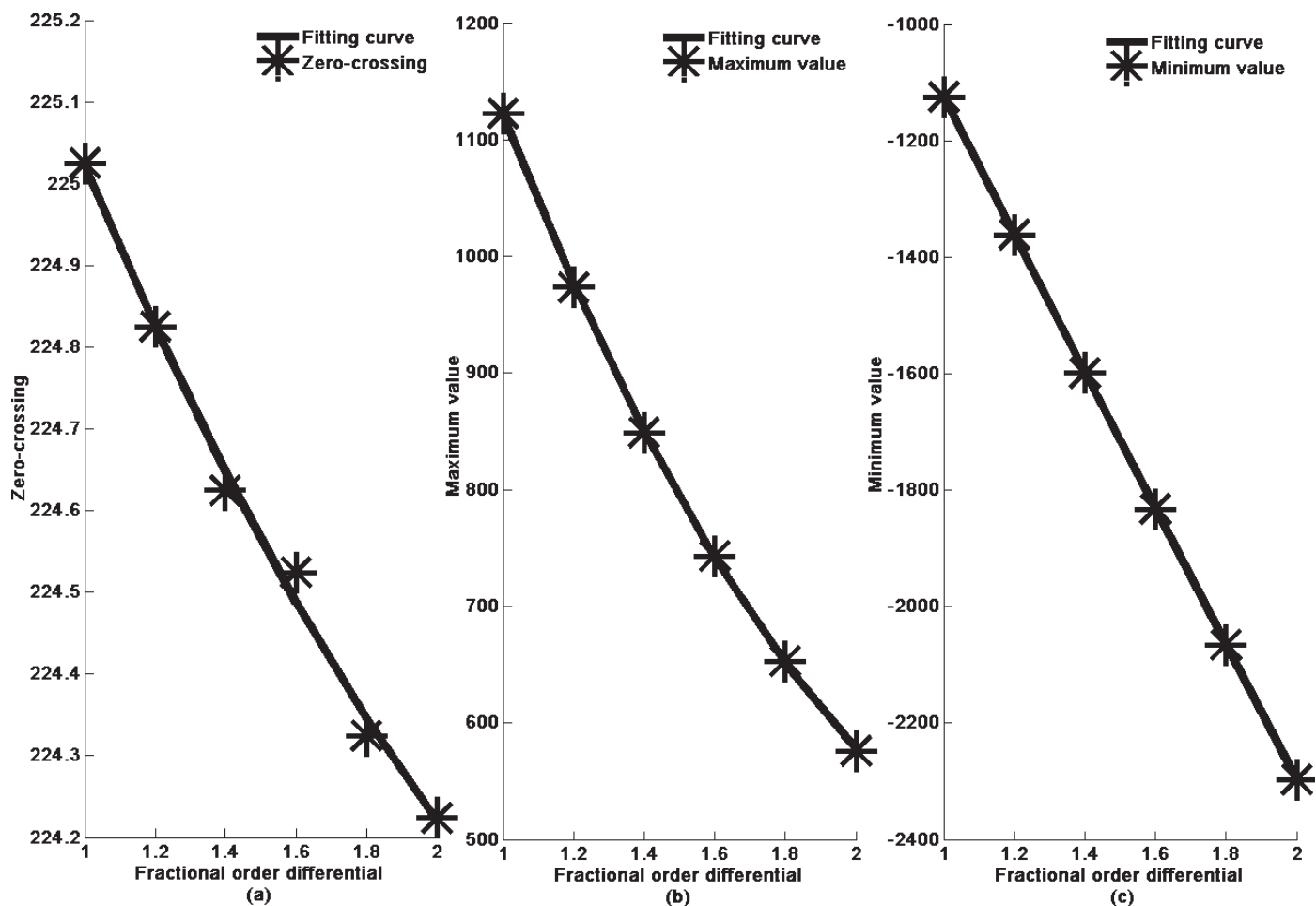


FIG. 2. Relationship of extremum and zero-crossing points with different fractional differential order α ($1 \leq \alpha \leq 2$) when $A = 2600.0$, $\mu = 225.0$, $\sigma = 1.5$, and $x \in [215, 235]$. (a) Relationship between zero-crossing points and the fractional differential order α . (b) Relationship between the maximum and the fractional differential order α . (c) Relationship between the minimum and the fractional differential order α .

where $x(\alpha)$ denotes the values of the α -order characteristic points, and a_2 , a_1 , and a_0 are the coefficients of the second-order multinomial. The relationship between the characteristic points and the corresponding differential order α is shown in Fig. 2, whose real lines denote fitting curves of the second-order multinomial models, and the asterisks denote the values of the α -order characteristic points. To be sure, the left sets of α -order differential local maximum and zero-crossing points were used to fit Eq. (5).

TABLE I. Parameter estimators of Lorentzian peaks.

Estimator I	$\begin{cases} \sigma_{estimated} = -\sqrt{3}(3a_2 + a_1) \\ \mu_{estimated} = a_2 + a_1 + a_0 \end{cases}$	Zero-crossing points
Estimator II	$\begin{cases} \sigma_{estimated} = \frac{0.7698(a_2 + a_1 + a_0)}{4a_2 + 2a_1 + a_0} \\ A_{estimated} = \frac{1.1852(a_2 + a_1 + a_0)^2}{4a_2 + 2a_1 + a_0} \end{cases}$	Local maximum
Estimator III	$\begin{cases} \sigma_{estimated} = \frac{3.0792(a_2 + a_1 + a_0)}{4a_2 + 2a_1 + a_0} \\ A_{estimated} = \frac{-4.7407(a_2 + a_1 + a_0)^2}{4a_2 + 2a_1 + a_0} \end{cases}$	Local minimum

According to Eq. (4) and Eq. (5), three types of parameter estimators are modeled for the Lorentzian peak. Estimator I is modeled by the zero-crossing points of α -order differential. Specifically, according to Eq. (4), we get $x(1) = \mu$ and $x(2) = (\sqrt{3}\mu \pm \sigma)/\sqrt{3}$. Inserting $x(1)$ and $x(2)$ into Eq. (5), Estimator I for μ and σ of Lorentzian peaks can be represent by

$$\begin{cases} \sigma_{estimated} = -\sqrt{3}(3a_2 + a_1) \\ \mu_{estimated} = a_2 + a_1 + a_0. \end{cases}$$

Similarly, Estimator II and Estimator III is modeled by corresponding local maximum and local minimum, respectively. The three-parameter estimators of Lorentzian peaks are listed in Table I. The $\sigma_{processed}$ is mutual in the three-parameter estimators. For unknown overlapped peaks, the validity of parameter estimation can be verified by $\sigma_{processed}$, which means three parameter estimators coincide.

Algorithms. The curve-fitting method requires extra procedures to reliably estimate the number and positions of the individual peaks to prevent ambiguous solutions. Considering the two parameters can be reliably determined by wavelet transform, a detailed procedure is not included in this study. The heights and FWHMs of the individual peaks were estimated by the parameter estimators of Lorentzian

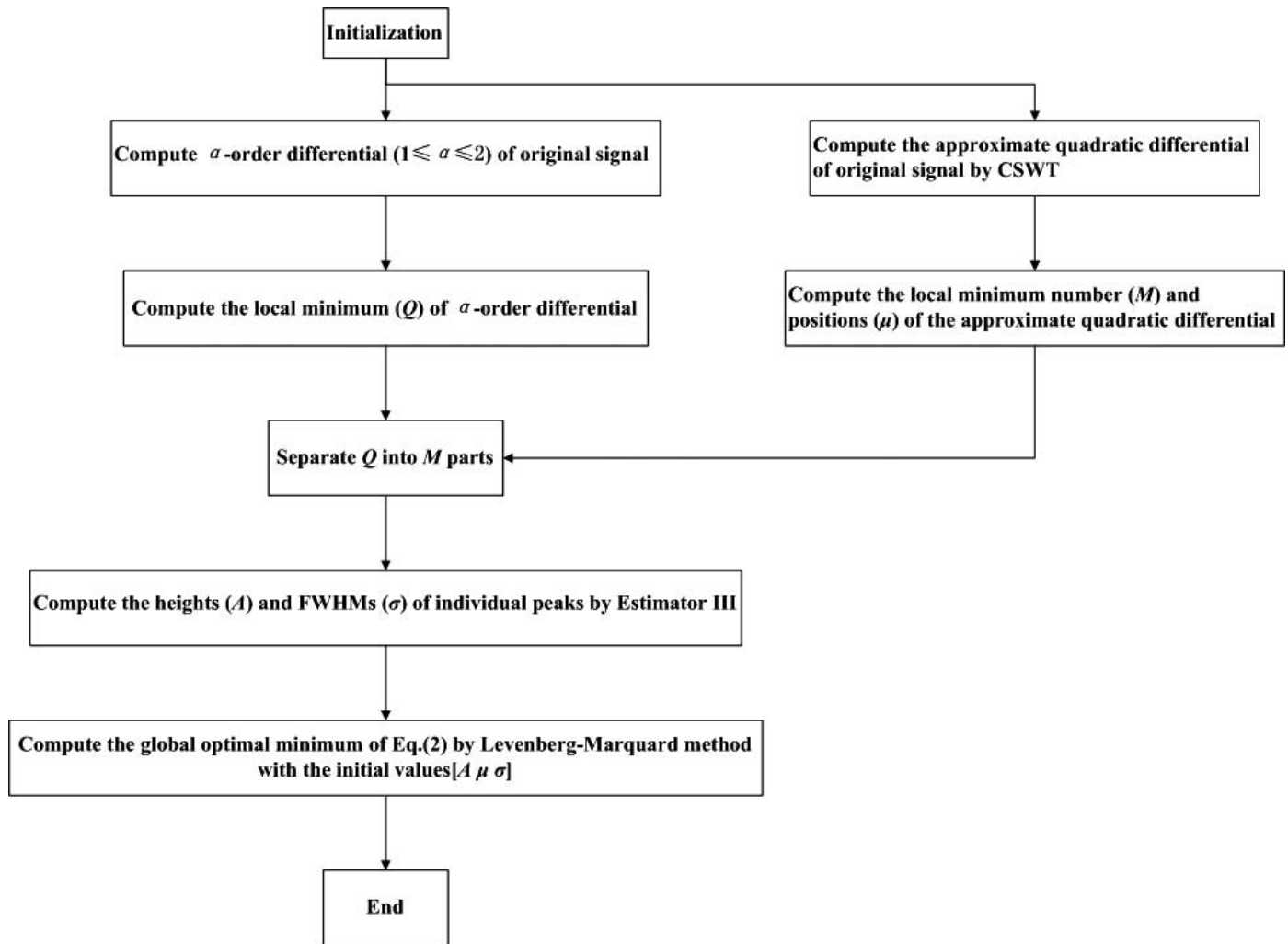


FIG. 3. Flow diagram of resolving overlapped peaks by Estimator III.

peaks. The key steps of resolving overlapped peaks are shown in Fig. 3.

According to the extreme value theorem, M and μ correspond to the number and positions of the individual peaks, respectively. In contrast, the μ of the peak positions also can be decided by Estimator I, which spends more time on grouping the zero-crossing points than the CSWT method. In addition, if we use Estimator II to get the heights and FWHMs of the individual peaks, then grouping local maximum is complicated and cumbersome. Therefore,

choosing Estimator III in the proposed method is simpler for the applications.

SIMULATION AND EXPERIMENT

To demonstrate the feasibility of the proposed method for resolving overlapped peaks, separation of several overlapped peaks shaped by Lorentzian models was performed. In addition, a LIBS experimental setup was prepared to test and verify the proposed method in practical applications.

TABLE II. Original and processed parameters, and error analysis.

	Original values			Processed values			Error analysis				
	A	μ	σ	$A_{processed}$	$\mu_{processed}$	$\sigma_{processed}$	ΔA	$\Delta \mu$	$\Delta \sigma$	RSS	$\mathfrak{R}$
$R = 0.25$	1	220	1	1.3699	220.2122	1.1193	0.3698	0.2112	0.1193	0.015917	0.9998
	1	221	1	0.6030	221.2179	0.7988	0.3970	0.2179	0.2011		
$R = 0.33$	1	220	1	1.0396	220.0235	1.0528	0.0396	0.0235	0.0528	0.002244	0.9999
	2	222	2	1.9661	222.0369	1.9801	0.0339	0.0369	0.0199		
$R = 0.5$	1	220	1	1.0005	220.0008	0.9997	0.0005	0.0008	0.0003	0.000004	0.9999
	1	222	1	0.9997	222.0007	0.9989	0.0003	0.0007	0.0011		
$R = 0.75$	1	220	1	1.0000	219.9997	0.9992	0.0000	0.0003	0.0008	0.000003	0.9999
	1	223	1	0.9998	222.9997	1.0008	0.0002	0.0003	0.0008		

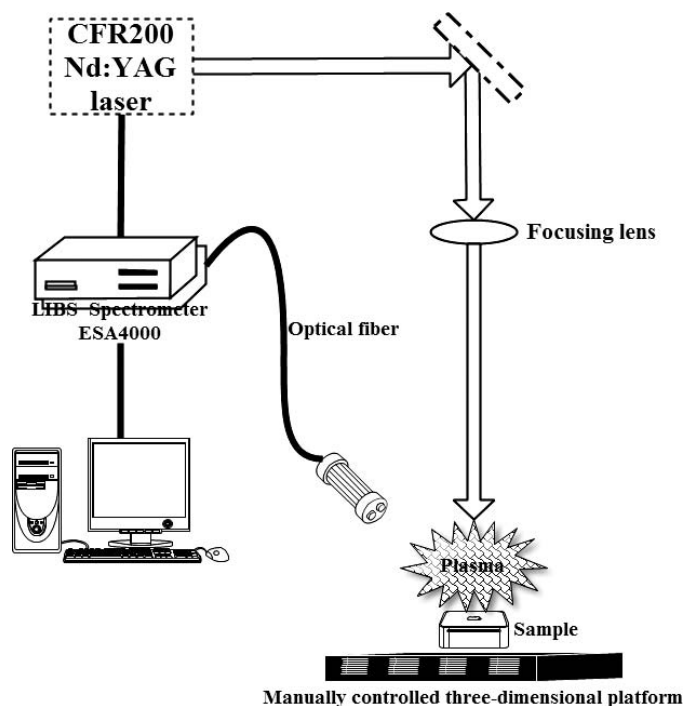


FIG. 4. Schematic visualization of the setup for the LIBS experiment.

Simulated Data. We consider the application of the proposed method for fitting noise-free overlapped peaks consisting of Lorentzian peaks. For convenience of expression, the separation degree is defined by $R = |\mu_i - \mu_j|/[2(\sigma_i + \sigma_j)]$, where μ_i and μ_j are the two peak positions and σ_i and σ_j are the FWHMs of two peaks.²⁵ The details of the simulated overlapped peaks with different parameters are listed in Table II.

LIBS Experiment. The LIBS experimental setup used for obtaining spectral data in this work is schematically shown in Fig. 4. The laser source was a Q-switched Nd:YAG laser (CFR200 Nd:YAG, Big Sky Laser) with maximum pulse energy of 200 mJ, pulse width of 12–15 ns, wavelength of 1064 nm, and repetition rate of 1–15 Hz. The sample was placed on a three-dimensional manually controlled stage, and the laser beam was focused on the sample surface by a convergent lens of 75 mm focal length. The spectrometer for analysis was a model ESA 4000 (LLA Instruments GmbH). In a wavelength range of 200–780 nm and with a resolution of a few picometers, spectra were recorded by an intensified charge-coupled device camera. The samples used in experiments were national standard Cu alloy samples (serial number GSB 04-2416-2008). For the measurements, the laser pulse energy was 100 mJ, the laser fluence on the sample was 50 J/cm², and the single laser pulse frequency was 5 Hz. Because LIBS data are influenced by a variety of experimental factors in the test process, the repeatability is low. Thus, we use the average of

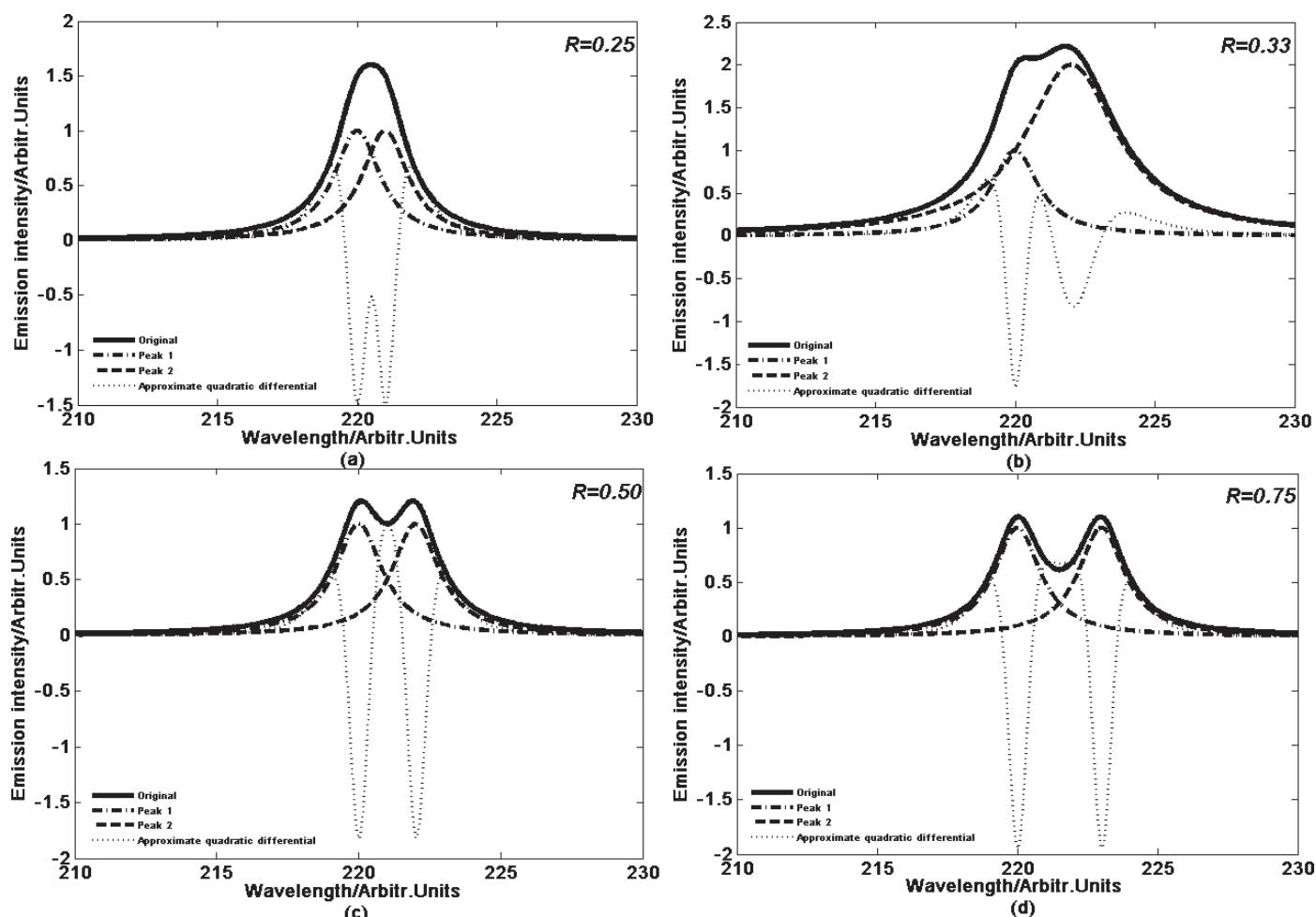


FIG. 5. Number and positions of the individual peaks obtained by CSWT.

multi-measurement to reduce the error caused by the uneven distribution of inner components. We measured each sample ten times in five different positions. All experiments were under the conditions of normal atmospheric pressure and 25 °C.

RESULTS AND DISCUSSION

Results of the Simulated Data. According to the feature of peak signal, several evaluation indices were chosen to evaluate the method's performance as follows: (1) the distortion of peak height by $\Delta A = |A - A_{processed}|$, where A is the original peak height and $A_{processed}$ is resolved peak height; (2) the difference of peak position by $\Delta \mu = |\mu - \mu_{processed}|$, where μ is the original peak position and $\mu_{processed}$ is resolved peak position; (3) the difference of FWHM by $\Delta \sigma = |\sigma - \sigma_{processed}|$, where σ is the original peak FWHM and $\sigma_{processed}$ is resolved peak FWHM; (4) the residual sum of squares (RSS) defined as Eq. (2); and (5) the correlation coefficients of fitness

$$\mathfrak{R} = \left(\sum XY - \frac{\sum X \sum Y}{N} \right) \div \sqrt{\left(\sum X^2 - \frac{(\sum X)^2}{N} \right) \left(\sum Y^2 - \frac{(\sum Y)^2}{N} \right)},$$

where X and Y is original data vector and curve-fitting data vector, respectively.

Figure 5 shows that even in a small separation degree, the number and positions of the individual peaks are clearly revealed by CSWT. As shown in Fig. 6, the parameters of the individual peaks are extracted accurately, which would be hard to detect directly. The error analysis of peak positions, peak heights, and FWHMs is shown in Table II.

According to the definition of separation degree, the peak positions and widths of two individual peaks are very important factors for the parameter estimation; thus, we explored the relationship of RSS and correlation coefficients of fitness with different separation degrees. The influence of the separation degree on the parameter estimation was observed, and the results are shown in Figs. 7 and 8. In Fig. 7, we can see that RSS decreases with increasing separation degree. In fact, RSS is kept constant when the separation degree exceeds a certain value. The results help us to explain why Fig. 6a has a larger RSS value than the other simulations. Similarly, Fig. 8 shows that as the separation degree increases, the correlation coefficients of fitness are close to 1. In fact, the peak heights have hardly any effect on correlation coefficients of fitness, and correlation coefficients approaches a steady state as the separation degree increases.

To compare with the results obtained by the proposed method and traditional method, Fig. 9 shows the processed

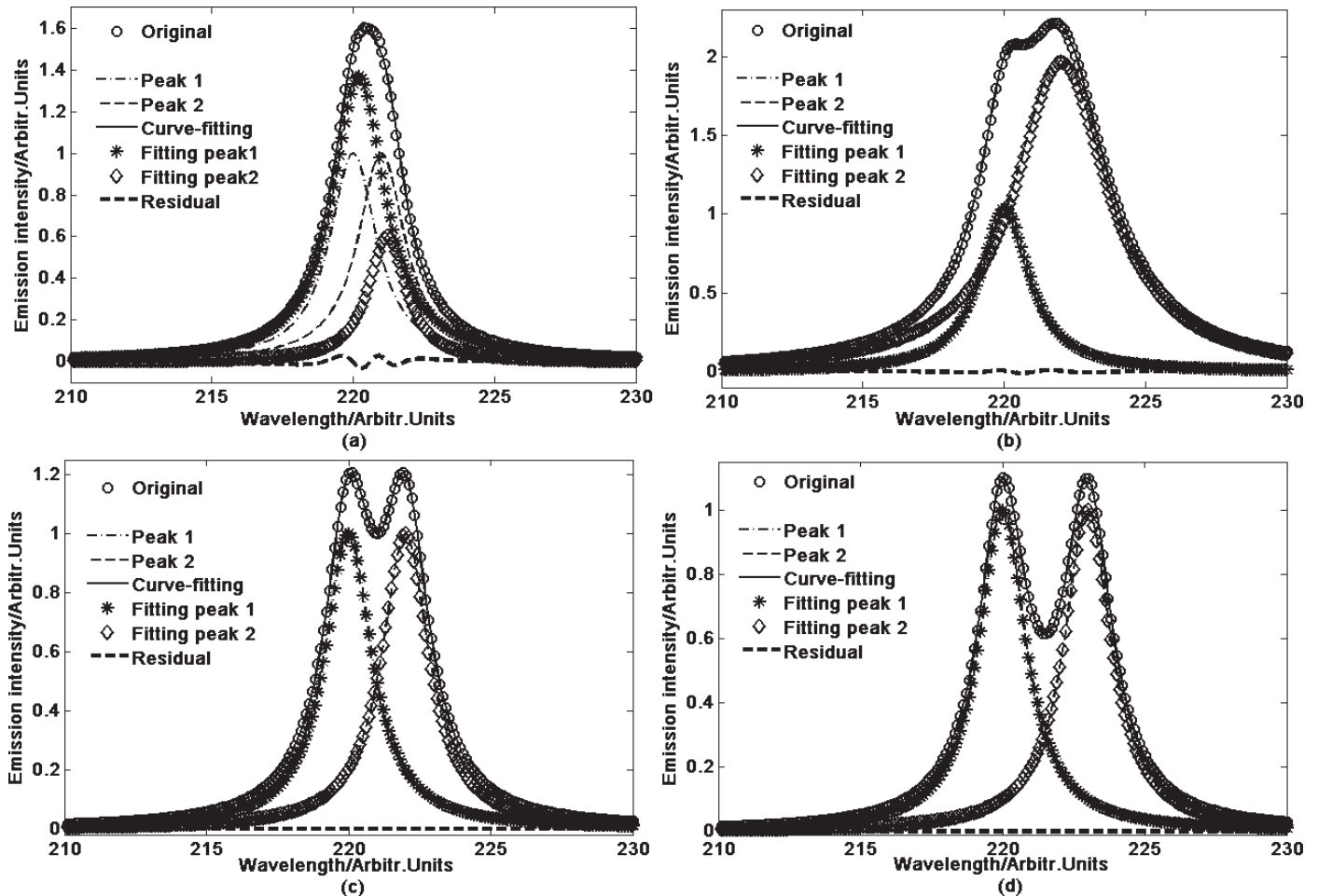


FIG. 6. Simulated overlapped peaks with different parameters were resolved by the proposed method.

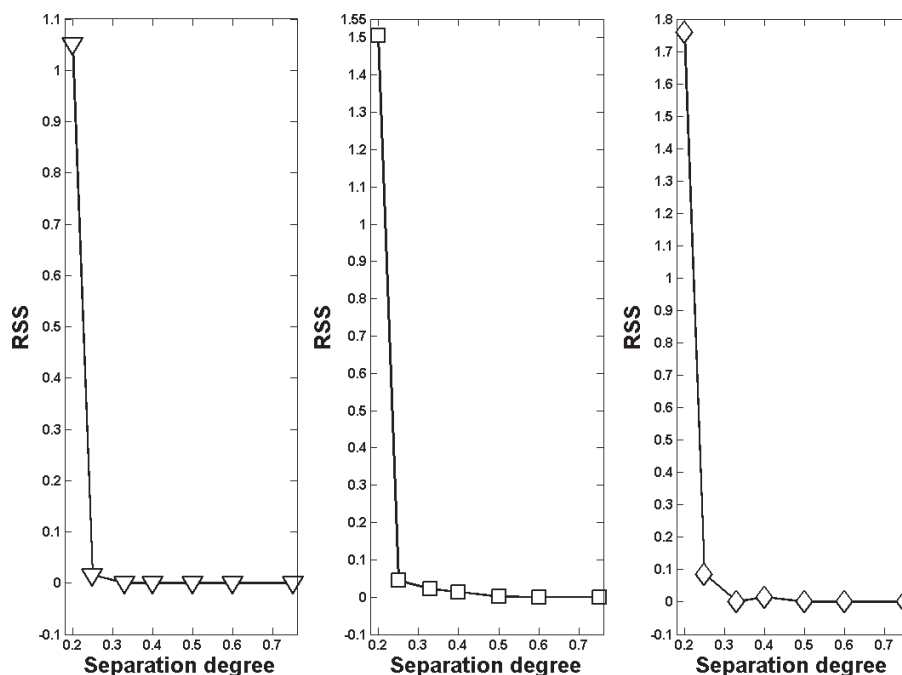


FIG. 7. Relationship between RSS and separation degree with different peak heights. From left to right, the peak heights are $(A_1, A_2) = (1,1)$, $(A_1, A_2) = (1,2)$, and $(A_1, A_2) = (1,3)$.

results of the overlapped Lorentzian peaks in Fig. 6b. The traditional method of the initial value selection usually proceeded as follows: (i) The number and positions of the individual peaks were fixed by the Fourier deconvolution method. (ii) All the peak heights were estimated by the local maximum of the overlapped peaks. (iii) All the widths of the overlapped peaks were fixed to a same estimated value. The values mentioned above were considered the initial values in the curve-fitting method. The details of Fig. 9 are listed in

Table III. Furthermore, the elapsed time of Fig. 9 is 14.11 s, whereas the elapsed time of Fig. 6b is 8.22 s. Since the comparison results are similar for the rest of Fig. 6, they are omitted here. The comparison results indicate that the proposed method is valid and reliable.

Results of the LIBS Experiment. With the purpose of verifying the effectiveness of the proposed method, quantification of some standard Cu samples was studied. The elemental concentrations of Cu samples are shown in Table IV.

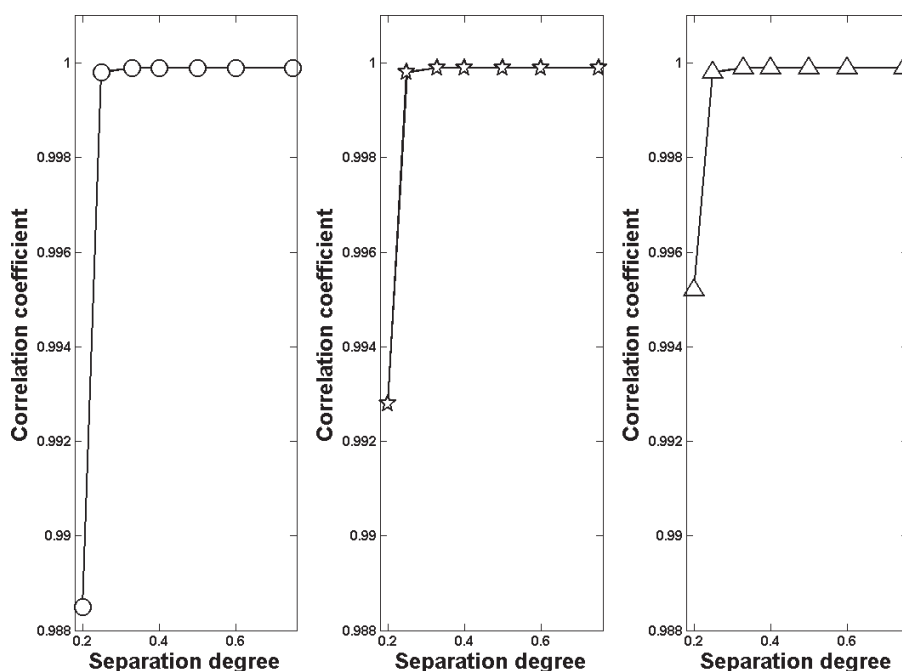


FIG. 8. Relationship between correlation coefficients of fitness and separation degree with different peak heights. From left to right, the peak heights are $(A_1, A_2) = (1,1)$, $(A_1, A_2) = (1,2)$, and $(A_1, A_2) = (1,3)$.

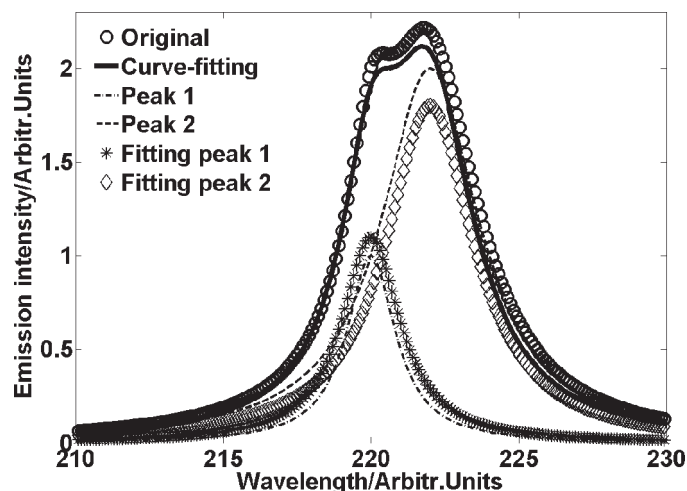


FIG. 9. Processed results of the overlapped Lorentzian peaks in Fig. 6b were obtained by the traditional method of the initial values' selection.

Since the measured data contained noise, pretreatment of the measured data was required. Specifically, we analyzed the signal-to-noise ratio (SNR), because the SNR refers to the ratio of the level of desired signal to the level of noise, and the higher the ratio, the less obtrusive the noise.²⁶ Moreover, the varying continuum background correction was accomplished by Lanxiang Sun's method.²⁷

TABLE III. Comparison results of different initial values estimated by different methods.

	Figure 6b		Figure 7	
Original values				
A	1	2	1	2
μ	220	222	220	222
σ	1	2	1	2
Initial values				
A	1.1713	1.8330	2.1014	2.2312
μ	220.120	222.122	219.697	221.798
σ	0.9857	1.7466	1.4	1.6
Processed values				
$A_{\text{processed}}$	1.0396	1.9661	1.1213	1.8423
$\mu_{\text{processed}}$	220.0235	222.0369	220.2014	222.6795
$\sigma_{\text{processed}}$	1.0528	1.9801	1.2252	1.8143
Error analysis				
ΔA	0.0396	0.0339	0.1213	0.1577
$\Delta \mu$	0.0235	0.0369	0.2014	0.6795
$\Delta \sigma$	0.0528	0.0199	0.2252	0.1857
RSS	0.002244		1.1148	
$\mathcal{R}$	0.9999		0.9788	

TABLE IV. Main elemental composition of six standard Cu samples used in resolving overlapped peaks.

Element	Concentration (%)					
	Cu alloy 1	Cu alloy 2	Cu alloy 3	Cu alloy 4	Cu alloy 5	Cu alloy 6
Cu	59.98	57.77	57.09	58.64	58.76	59.6
Fe	0.205	0.0426	0.0104	0.703	0.107	0.613
Ni	1.474	0.795	0.347	0.104	0.00286	0.386
Al	0.0749	0.0116	0.177	0.452	0.761	0.262

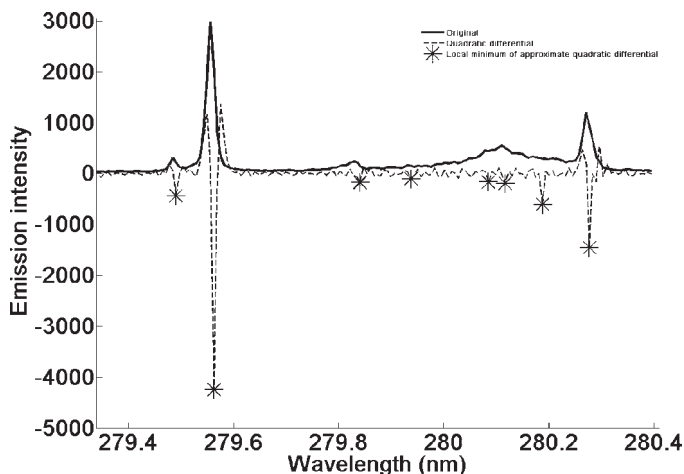


FIG. 10. Processed results of the number and positions obtained by CSWT.

As shown in Fig. 10, the processed results of the estimated number and positions of individual peaks were obtained by CSWT. Based on finding negative regions in the approximate quadratic differential of the original signal, the number and positions of the individual peaks were decided by extreme value theorem. However, not all negative regions are due to the efficient signal. Peaks also can be due to noise. To account for this noise, the noise in the approximate quadratic differential

TABLE V. Results of resolved overlapped peaks for Fig. 16.

Peak no.	Wavelength (nm) from ASD	$A_{\text{processed}}$	$\mu_{\text{processed}}$	$\sigma_{\text{processed}}$	RSS	$\mathcal{R}$
1	279.55398 (Fe)	2713	279.5630	0.0021	22.1430	0.9889
2	280.310 (Fe)	1084	280.2770	0.0062		
3	279.482 (Mn)	186	279.4910	0.0070		
4	280.1180 (Al)	249	280.1170	0.0173		
5	279.8473 (Ni)	173	279.8410	0.0091		
6	—	168	280.1870	0.0854		
7	280.087 (Zn)	250	280.0850	0.0432		
8	279.9528 (Cu)	40	279.9380	0.0636		

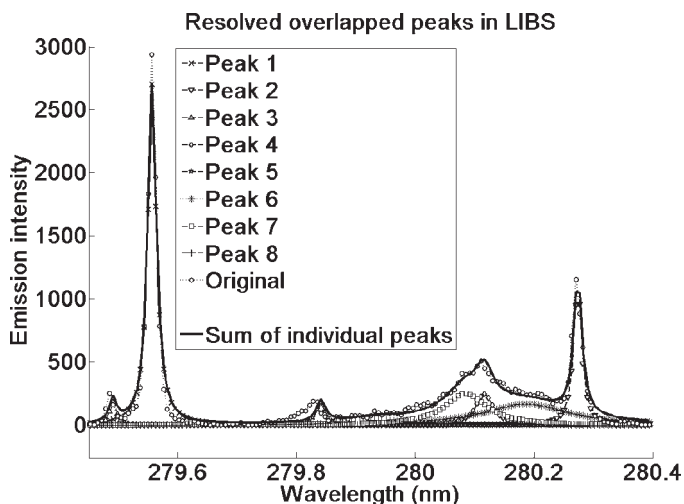


FIG. 11. Resolved overlapped peaks in the wavelength range of 279.3–280.4 nm by the proposed method.

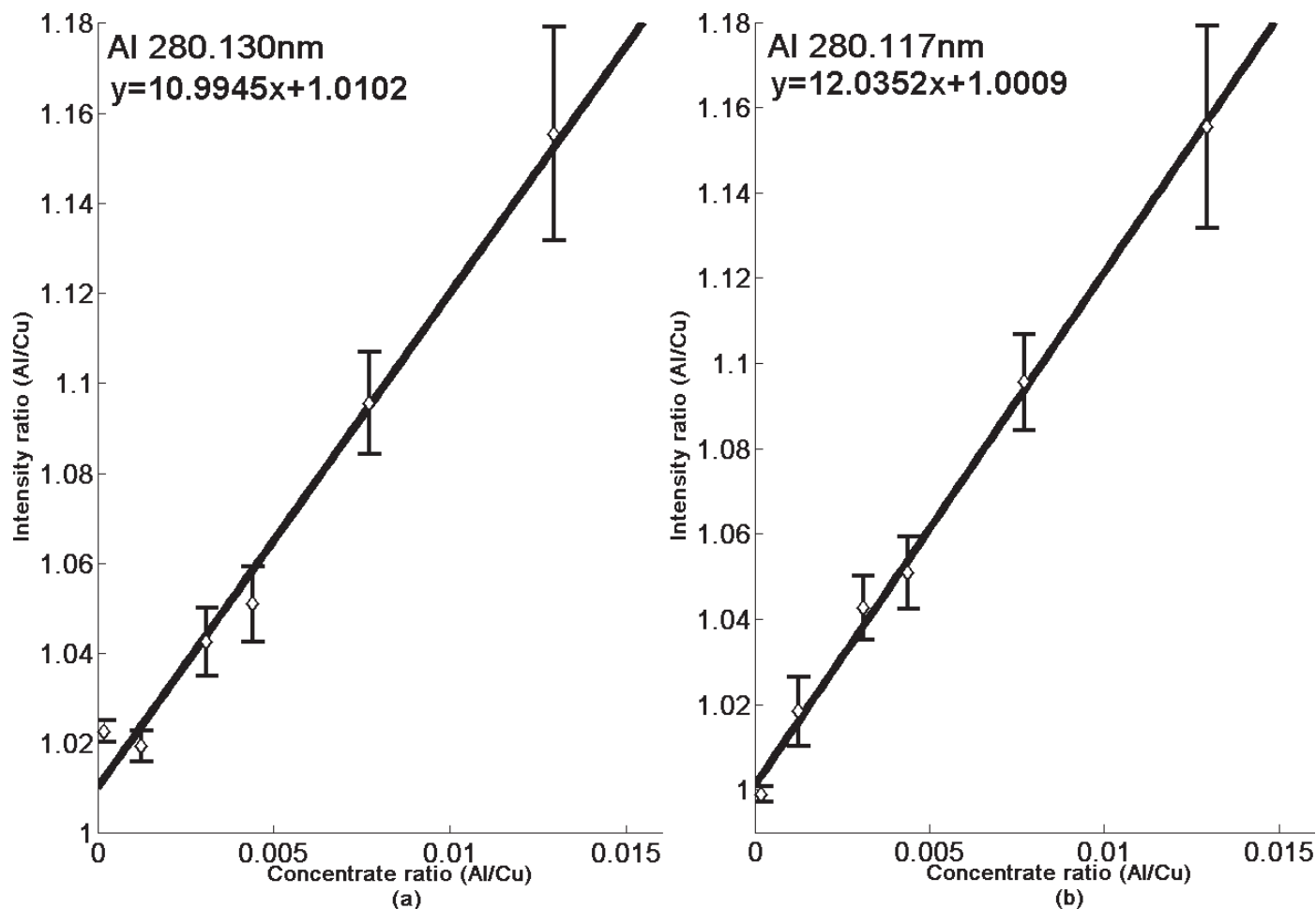


FIG. 12. Calibration curves for Al. Cu 374.469 nm was chosen as the internal standard line. (a) Before resolving overlapped peaks, at 280.130 nm, the correlation coefficient was 0.9857 and the relative standard deviation was 0.0173. (b) By using the proposed method of the initial values' estimation for curve fitting, at 280.117 nm, the correlation coefficient was 0.9964 and the relative standard deviation was 0.0168.

was computed to establish a cut-off value that allowed distinguishing real peaks from noise.^{28,29} Table V and Fig. 11 show the detailed results of eight individual peaks in the overlapped peak.

The elements identified of the individual peaks need to be determined by comparison with the National Institute of Standards and Technology Atomic Spectra Database (ASD). Unfortunately, we cannot find the position of peak 6 corresponding to the wavelength in the ASD. The phenomenon is subject to accumulation errors, including random noise and the continuum background in the spectral. Figure 12 gives a comparison of calibration curves results for Al before and after resolving the overlapped peaks. The results demonstrate that the overlapped peaks with low separation degree have been effectively separated by the proposed method. The fitting results show that the errors in spectral parameter estimation are due mainly to accumulation of errors involving random noise and continuum background.

To evaluate the effect of the proposed method, we compared our results with results obtained by different methods of the initial value estimation in curve fitting. Two individual peaks were resolved in the overlapped peaks by the proposed method in Fig. 13. The fitting errors in Table VI indicate that the best resolving results were obtained by the proposed method. Figure

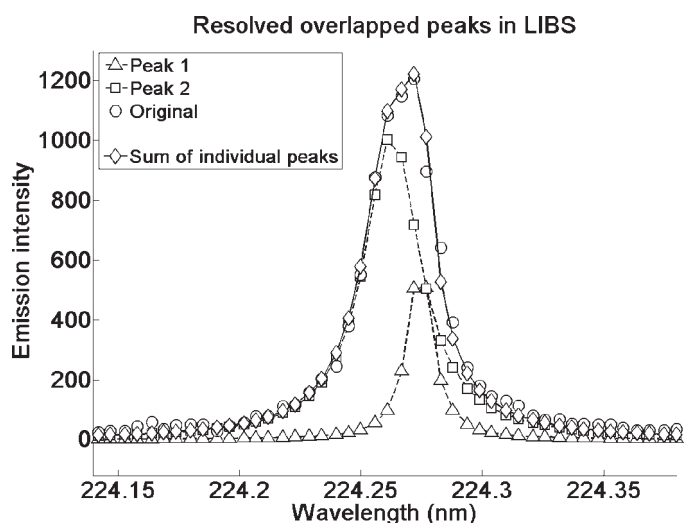


FIG. 13. Resolved overlapped peaks in the wavelength range 224.14–224.40 nm by the proposed method.

TABLE VI. Results of resolving overlapped peaks for Fig. 14.

	Peak no.	Wavelength (nm) from ASD	$A_{processed}$	$\mu_{processed}$	$\sigma_{processed}$	RSS	$\mathfrak{R}$
Traditional method	1	224.27807 (Fe)	585	224.2801	0.0061	28.3152	0.9731
	2	224.2618 (Cu)	738	224.2689	0.0120		
Proposed method	1	224.27807 (Fe)	548	224.2776	0.0046	25.3014	0.9965
	2	224.2618 (Cu)	1021	224.2616	0.0117		

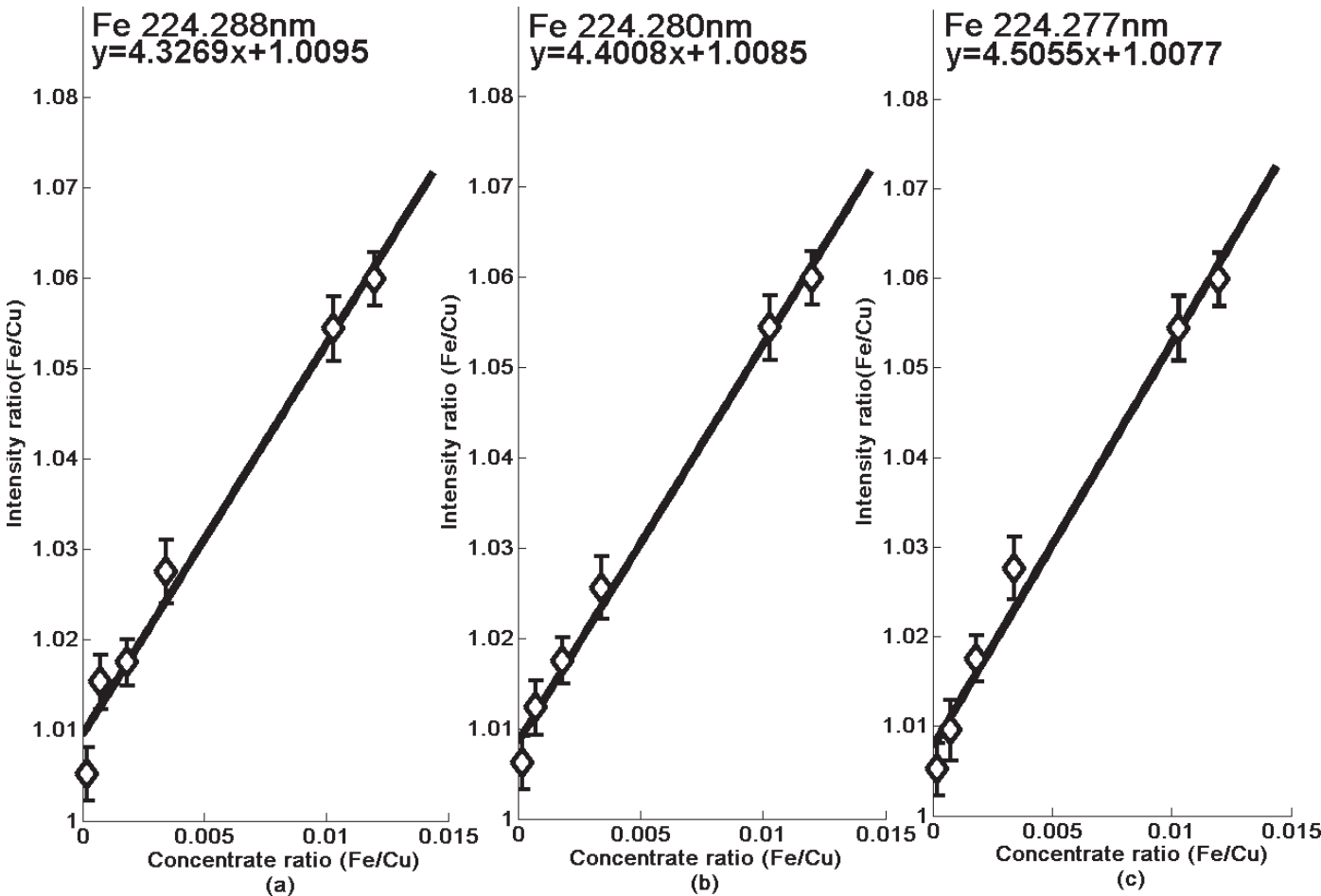


FIG. 14. Calibration curves for Fe. Cu 374.469 nm was chosen as the internal standard line. (a) Before resolving overlapped peaks, at 224.288 nm, the correlation coefficient was 0.9813 and the relative standard deviation was 0.0170. (b) By using the traditional method of the initial values' estimation for curve fitting, at 224.280 nm, the correlation coefficient was 0.9857 and the relative standard deviation was 0.0154. (c) By using the proposed method of the initial values' estimation for curve fitting, at 224.277 nm, the correlation coefficient was 0.9932 and the relative standard deviation was 0.0122.

14 gives a comparison of calibration curves results for Fe before and after resolving the overlapped peaks. The quality of the calibration curves built on resolving individual peaks is better than that built on the original overlapped peaks in sensitivity and correlation coefficients.

CONCLUSIONS

In this work, we have developed parameter estimators based on the fractional differential theory to resolve overlapped peaks in LIBS data. With the help of parameter estimators, the initial values of curve-fitting methods were selected accurately. According to the preliminary investigation, we successfully proved the efficiency and accuracy of the proposed method

from the results of the simulations and experiments. The proposed method is not only a valid tool of resolving overlapped peaks in LIBS but also a useful tool in other spectral signals.

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